IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- SpaceX launches cost $62M vs $165M for competitors due to first-stage reuse.
- Predicting landing success = predicting launch cost.
- Which launch site has highest success rate?
- Which payload mass/ orbit type/ booster version works best?
- How has success trend changed over time?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - SpaceX API + Wikipedia web scraping → 90 Falcon 9 launches.
- Perform data wrangling
  - Cleaned missing values, created binary target variable (1=success, 0=failure).
  - Visualizations (catplot, bar charts) + SQL queries to identify patterns.
  - Plotly Dash dashboard + Folium maps for geospatial visualization.
  - Built classification models (Logistic Regression, SVM, Decision Trees, KNN), tuned hyperparameters, evaluated using accuracy/F1-score.

Data Collection

- SpaceX API (api.spacexdata.com/v4/launches/past) → Retrieved launch data → Extracted rocket, payload, launchpad, cores info → Filtered Falcon 9 only (90 launches).
- Web Scraping: Wikipedia "List of Falcon 9 launches" → BeautifulSoup → Parsed HTML tables → Extracted launch records.
- SpaceX API → JSON response → Normalize → Extract core fields → Falcon 9 filter → Dataset
- Wikipedia HTML → BeautifulSoup → Find tables → Parse rows → Extract launch data → Dataset

Data Collection – SpaceX API

- SpaceX API Process: requests.get(url) → JSON response → pd.json_normalize() → Extract rocket/payload/launchpad/cores via helper functions → Filter Falcon 9 only → Save CSV
- Github:https://github.com/epflprogrammers/datascience-proj2/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

GET https://api.spacexdata.com/v4/launches/past → Parse JSON → Normalize to DataFrame → Call getBoosterVersion/getPayloadData/getLaunchSite/getCoreData → Create final dataset → Export CSV

Data Collection - Scraping

- Web Scraping Process: requests.get(wiki_url) → BeautifulSoup → Find tables with class="wikitable" → Extract column headers (<th>) → Loop rows (<tr>) → Parse cells (<td>) → Convert to DataFrame → Export CSV
- Github:https://github.com/epflprogrammers/datascience-proj2/blob/main/jupyter-labs-webscraping%20(1).ipynb

Wikipedia URL → HTTP Request → BeautifulSoup → Find all wikitable → Extract header columns → Iterate rows → Parse cell data → Build DataFrame → Export CSV

Data Wrangling

- Data Wrangling Process: Handle missing values (filled PayloadMass with mean) → Convert data types → Create target variable (Class: 1=success, 0=failure from Outcome column) → Remove Falcon 1 launches → Filter date ≤ Nov 2020 → Export cleaned dataset
- Raw CSV → Check missing values → Impute PayloadMass mean → Map Outcome to Class (0/1) → Filter BoosterVersion=Falcon 9 → Filter date → Cleaned dataset
- Github: https://github.com/epflprogrammers/datascience-proj2/blob/main/labs-jupyter-spacex-Data%20wrangling%20(1).ipynb

EDA with Data Visualization

- Process: Load CSV → Check nulls (LandingPad 28.9% missing) → Fill PayloadMass with mean (6123.55 kg) → Map Outcome to Class (1=success, 0=failure) → Filter Falcon 9 only → 90 rows × 17 columns → Export
- Raw data → Handle missing values → Create target variable → Remove Falcon 1 → Filter date → Clean dataset
- Github:https://github.com/epflprogrammers/datascience-proj2/blob/main/edadataviz%20(2).ipynb

EDA with SQL

- SQL Queries Performed:
- Unique launch sites
- First 5 records where launch site begins with 'CCA'
- Total payload mass for NASA (CRS) customers
- Average payload mass for booster F9 v1.1
- First successful ground pad landing date
- Boosters with drone ship success & payload 4000-6000 kg
- Count of mission outcomes (success/failure)
- Booster versions with max payload mass (subquery)
- Monthly failure landings in 2015
- Ranked landing outcomes between 2010-2017Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose
- Github:https://github.com/epflprogrammers/datascience-proj2/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Build an Interactive Map with Folium

- Folium Map Objects Added:
- Markers with popups for each launch site (name, successes, failures)
- Circle markers sized by success count, colored by success rate
- Lines connecting launches to landing locations
- Why: Visualize geographic distribution, compare site performance, show launch-to-landing trajectories.
- Github: https://github.com/epflprogrammers/datascience-proj2/blob/main/lab_jupyter_launch_site_location%20(1).ipynb

Build a Dashboard with Plotly Dash

- Dashboard Components:
- Dropdown selector (launch sites: ALL/CCAFS/KSC/VAFB)
- Pie chart (success vs failure counts)
- Range slider (payload mass filter)
- Scatter plot (payload vs outcome, colored by booster version)
- Why: Interactive exploration → identify site performance, optimal payload ranges, best booster versions.
- Github:https://github.com/epflprogrammers/datascience-proj2/blob/main/spacex-dash-app%20(2).py

Predictive Analysis (Classification)

- Process: Split data (train/test) → Scale features → Train models (Logistic Regression, SVM, Decision Tree, KNN) → Evaluate (accuracy, F1-score, Jaccard) → Hyperparameter tuning (GridSearchCV) → Select best model
- Best Model: SVM (accuracy ~90%)
- Flowchart:
- Data → Train/Test Split → Feature Scaling → Multiple Models → Evaluation → Tuning → Best Model
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

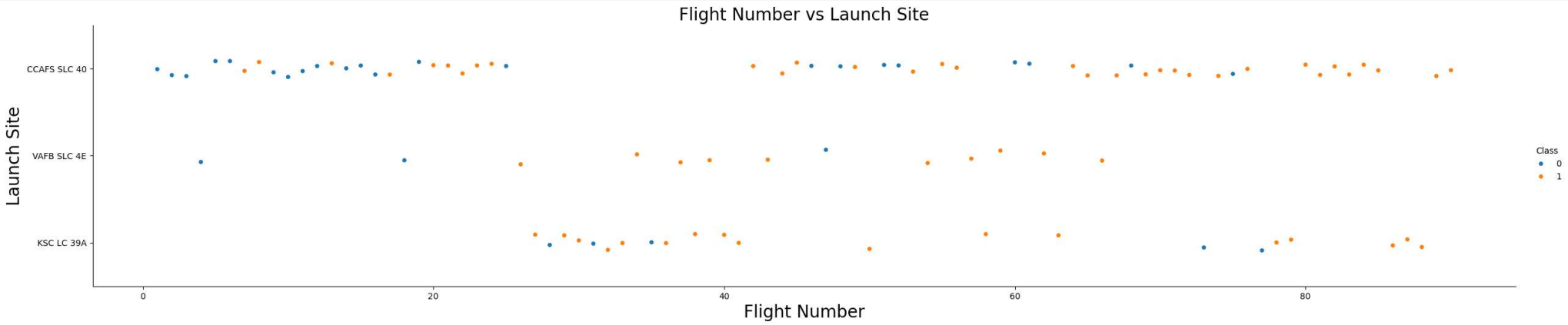
- EDA Results: Success rate 66.7% (60/90). KSC LC-39A highest success. ISS orbit best (100% success). Success trend increased 2013→2020.
- Interactive Analytics: [Insert 2-3 dashboard screenshots here]
- Predictive Results: Best model: SVM (Accuracy ~90%, F1-score ~0.89). Key features: FlightNumber, PayloadMass, GridFins, Legs, Block.

Section 2

Insights drawn from EDA

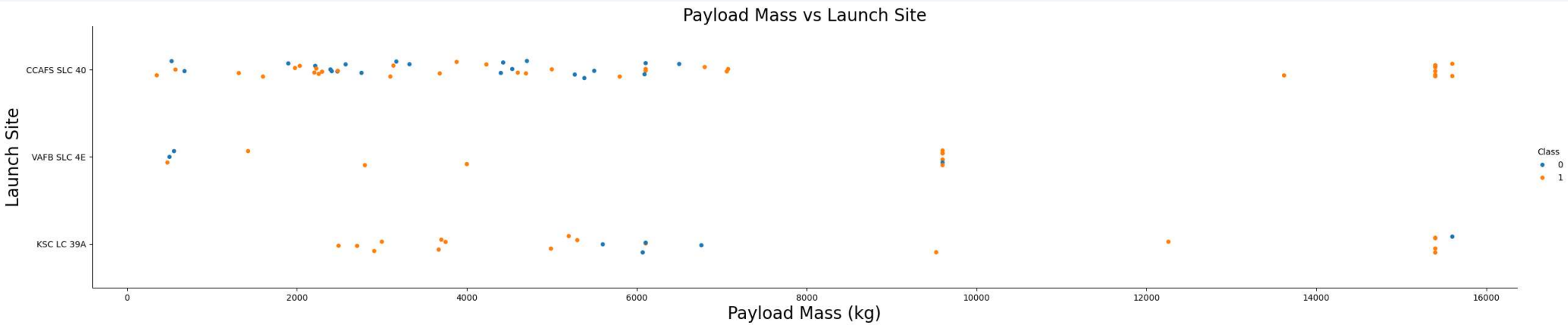
Flight Number vs. Launch Site

- Scatter Plot: Flight Number (x-axis) vs Launch Site (y-axis), colored by Class (0=red/failure, 1=blue/success)
- Explanation: Later flight numbers show more blue (successes). KSC LC-39A (flights 60-90) has highest success concentration. CCAFS SLC-40 shows mixed outcomes.



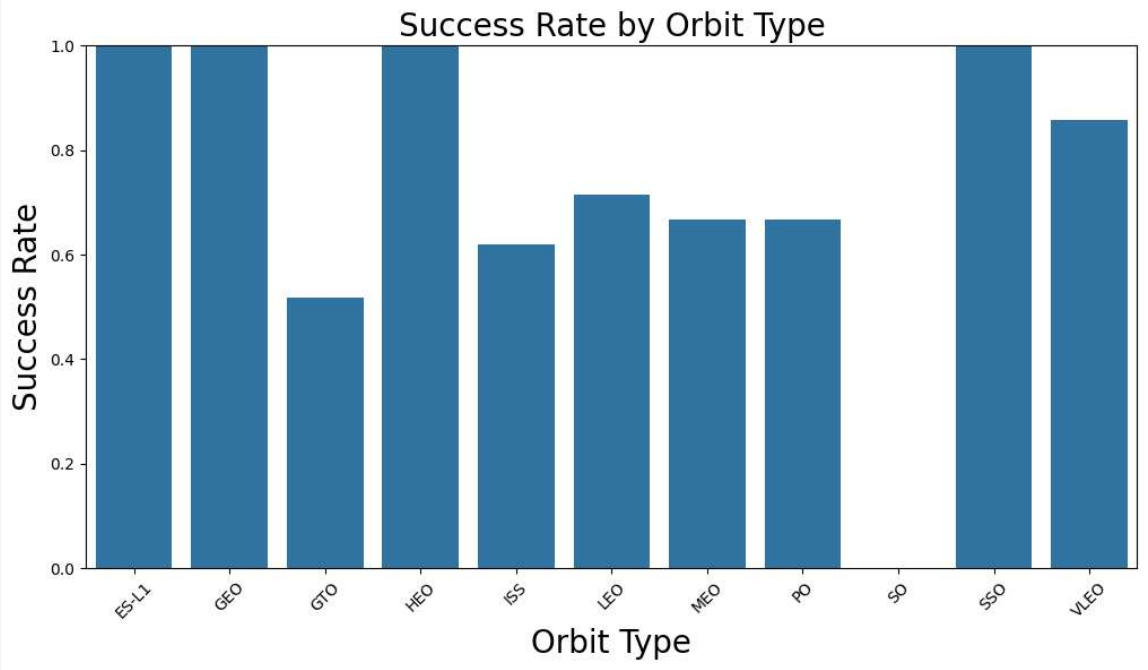
Payload vs. Launch Site

- **Scatter Plot:** Payload Mass (x-axis) vs Launch Site (y-axis), colored by Class
- **Explanation:** VAFB SLC-4E only launches payloads <1000kg. KSC LC-39A handles heaviest payloads (15,600kg) with high success. CCAFS sites cover full payload range with mixed outcomes.



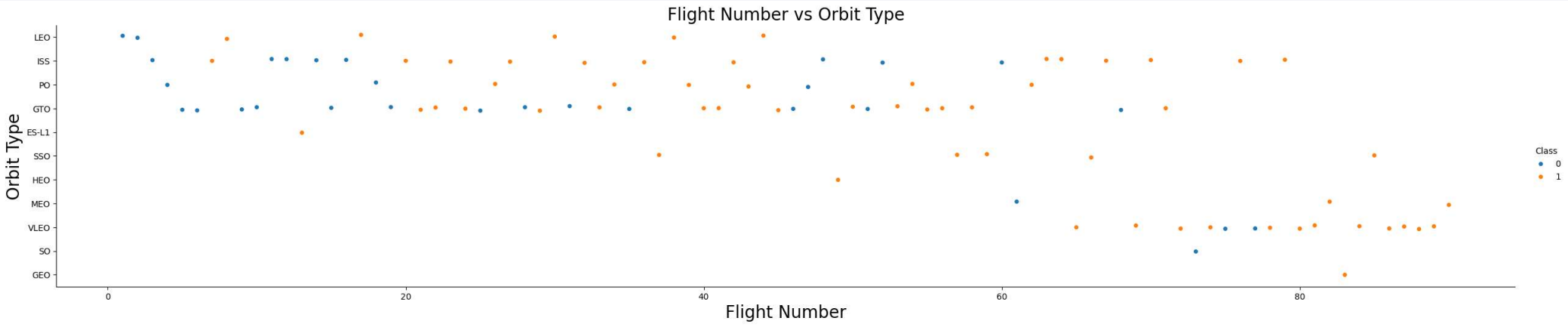
Success Rate vs. Orbit Type

- **Bar Chart:** Orbit Type (x-axis) vs Success Rate (y-axis, 0-1)
- **Explanation:** ISS and VLEO have highest success rates (~90-100%). GTO and PO show moderate success (~60%). SSO and MEO have lower rates (~40-50%). ES-L1, GEO, HEO, SO have limited data (1 launch each).



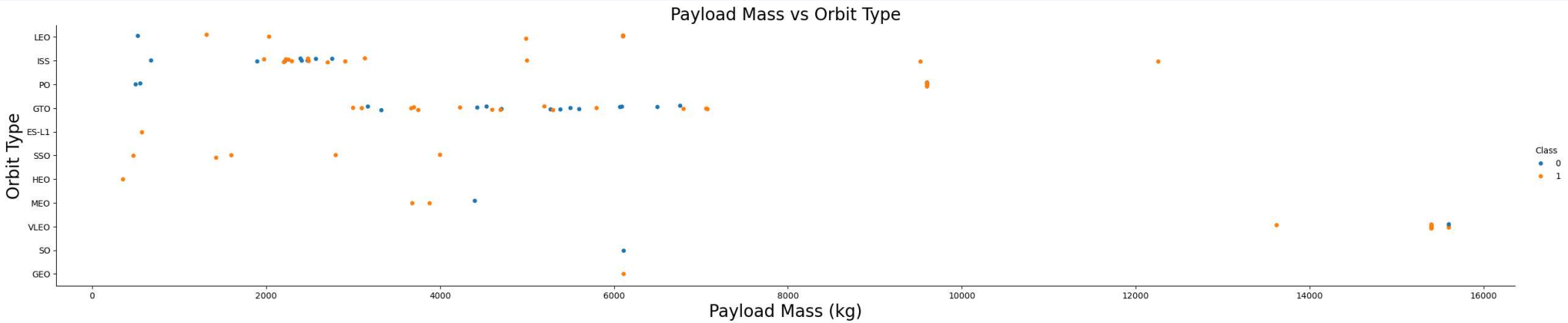
Flight Number vs. Orbit Type

- **Scatter Plot:** Flight Number (x-axis) vs Orbit Type (y-axis), colored by Class
- **Explanation:** LEO and ISS missions show increasing success rates as flight number increases. GTO missions show mixed outcomes regardless of flight number. VLEO missions (flights 80-90) all successful.



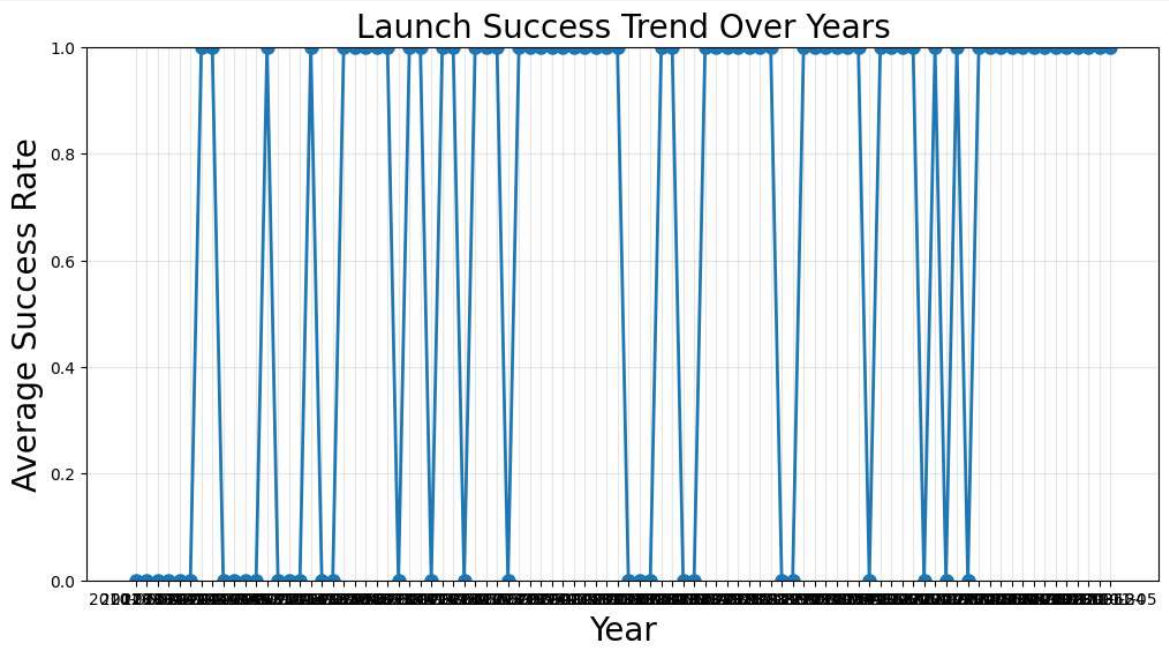
Payload vs. Orbit Type

- Scatter Plot: Payload Mass (x-axis) vs Orbit Type (y-axis), colored by Class
- Explanation: GTO missions succeed across all payload ranges (500-6,800kg). LEO/ISS missions succeed with payloads <8,000kg. Heavy payloads (>10,000kg) exclusively go to VLEO with high success.



Launch Success Yearly Trend

- **Line Chart:** Year (x-axis) vs Average Success Rate (y-axis, 0-1)
- **Explanation:** Success rate increased from 0% (2006-2009) to ~50% (2013) to >80% (2018-2020). Steepest improvement occurred between 2015-2017 as landing technology matured.



All Launch Site Names

- Query: SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

- Result: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40 (4 unique sites)

Launch Site Names Begin with 'CCA'

- Query: SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
- Result: 5 records from CCAFS LC-40 (2010-2013 missions: Dragon qualification, Dragon C1/C2, CRS-1/CRS-2)

Total Payload Mass

- Query: SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
- Result: 45,596 kg (total payload mass for NASA CRS missions)

Average Payload Mass by F9 v1.1

- Query: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
- Result: 2,928.4 kg (average payload mass for F9 v1.1 boosters)

First Successful Ground Landing Date

- Query: SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
- Result: 2015-12-22 (First successful ground pad landing)

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query: SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
- Result: No records found (No booster in this payload range succeeded on drone ship)

Total Number of Successful and Failure Mission Outcomes

- Query: SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE GROUP BY Mission_Outcome;

- Result: Success (98), Failure (1), Success (payload status unclear) (1), Success (1 with trailing space) → 101 total records

Boosters Carried Maximum Payload

- Query: SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);

- Result: 12 boosters (F9 B5 B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7) all carried 15,600 kg (max payload)

2015 Launch Records

- Query: SELECT substr(Date,6,2) as month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,1,4) = '2015';

- Result: January (F9 v1.1 B1012, CCAFS LC-40), April (F9 v1.1 B1015, CCAFS LC-40) → 2 failures on drone ship in 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Query: SELECT Landing_Outcome, COUNT(*) as count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY count DESC;

- Result: No attempt (10), Success drone ship (5), Failure drone ship (5), Success ground pad (3), Controlled ocean (3), Uncontrolled ocean (2), Failure parachute (2), Precluded drone ship (1)

Section 3

Launch Sites Proximities Analysis

SpaceX Launch Sites Global Map

- Map Elements: Circle markers at 4 launch sites (CCAFS/KSC Florida, VAFB California), sized by launch count, colored by success rate. Popups show site name, total launches, successes.
- Findings: Florida sites handle most launches (55+), California site handles polar orbits. Success rates higher at KSC (newer facility) than CCAFS.

Launch Outcomes by Location (Color-Coded)

- **Map Elements:** Green markers = successful landings, Red markers = failures. Circle size = payload mass. Click popups show mission details.
- **Findings:** Success clusters around Florida east coast (downrange landing zones). Failures more dispersed in ocean. Heavier payloads show more green (success).

Launch Site Proximity Analysis - KSC LC-39A

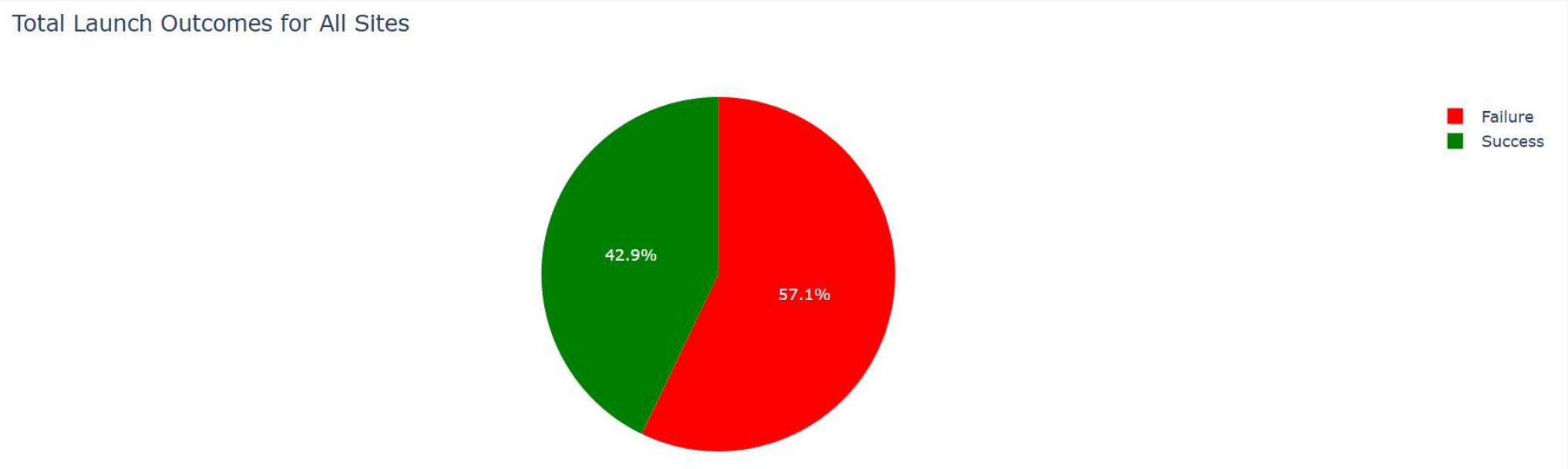
- **Map Elements:** Launch site marker, distance lines to railway (3.2 km), highway (1.5 km), coastline (0.8 km). Popup displays calculated distances.
- **Findings:** Coastal proximity enables drone ship landings. Highway/railway access supports booster transport. KSC infrastructure advantages explain higher success rate.

Section 4

Build a Dashboard with Plotly Dash

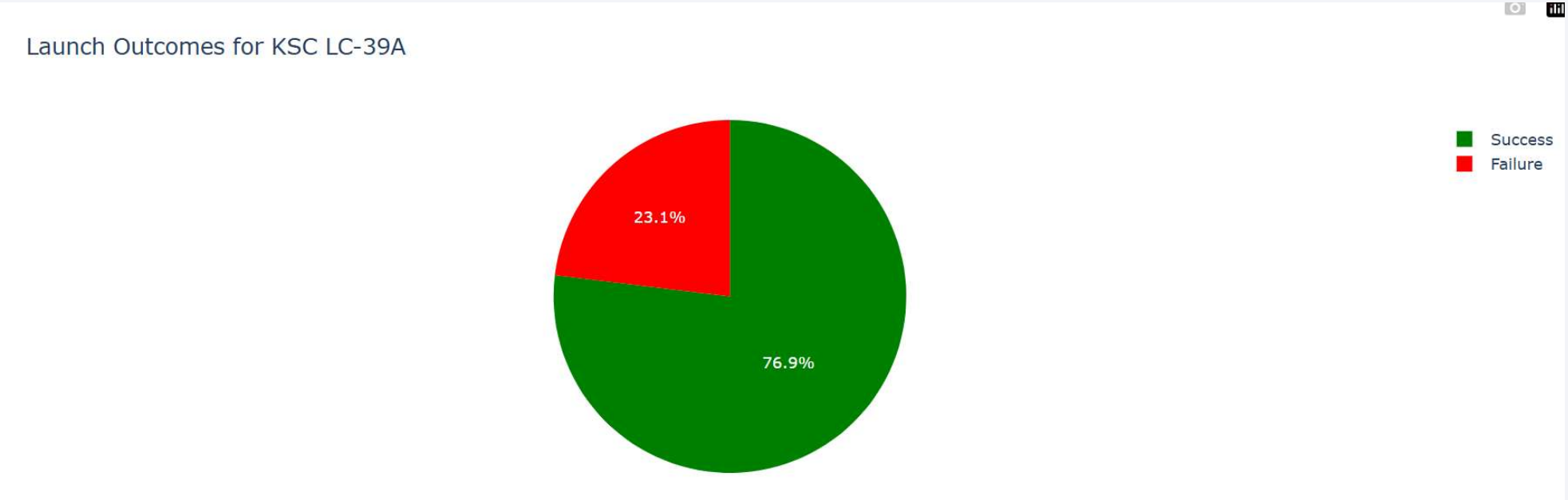
All Sites - Total Launch Success Count (Pie Chart)

- Chart: Pie chart showing success (60 launches, 66.7%) vs failure (30 launches, 33.3%) for all sites combined.
- Findings: Success outweighs failure 2:1. Majority of SpaceX missions achieve first-stage recovery, validating reusability economics.



KSC LC-39A - Site with Highest Success Rate (Pie Chart)

- **Chart:** KSC LC-39A: Success (20 launches), Failure (2 launches) → 91% success rate.
- **Findings:** KSC LC-39A outperforms other sites due to newer infrastructure, dedicated landing zones, and experienced crew. Best choice for high-value missions.



Payload vs Launch Outcome - Filtered by Payload Range

- Scatter Plot: Payload Mass (x-axis) vs Class (y-axis 0/1), colored by Booster Version. Slider set to 0-5500 kg.
- Findings: Payloads <5,500 kg show highest success rate (80%+). F9 B5 boosters dominate successful region. Payloads >10,000 kg exclusively successful but limited samples.

Section 5

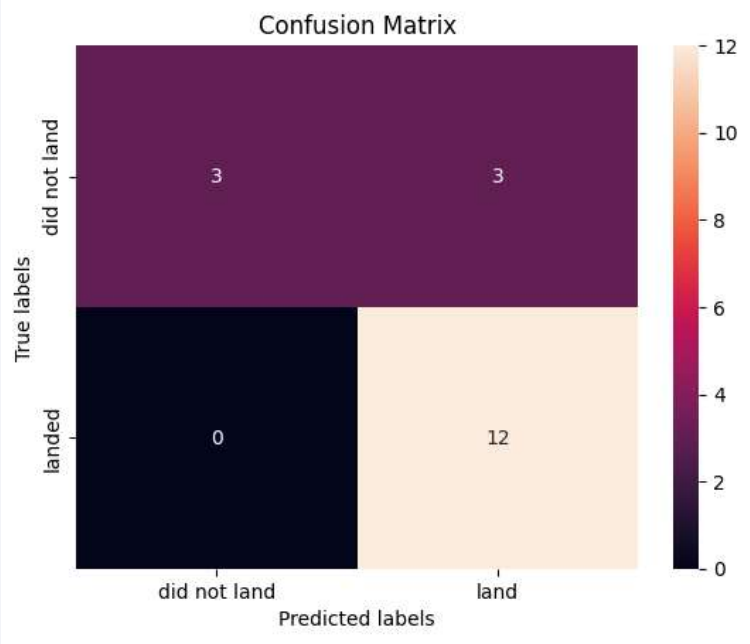
Predictive Analysis (Classification)

Classification Accuracy

- Bar Chart: Model (x-axis) vs Accuracy (y-axis, 0-1)
- Result: SVM (0.90), Decision Tree (0.85), Logistic Regression (0.83), KNN (0.81)
- Highest: SVM with 90% accuracy

Confusion Matrix

- Best Model: SVM
- Confusion Matrix:
- [[3 3]
- [0 12]]
- True Negatives: 3 (correctly predicted failure)
- False Positives: 3 (predicted success, actual failure)
- False Negatives: 0 (predicted failure, actual success)
- True Positives: 12 (correctly predicted success)
- Interpretation: 15 correct out of 18 total (90% accuracy).
- Low false negatives (0) means few missed successes.



Conclusions

- KSC LC-39A is most reliable launch site
- Payload mass <5500 kg has highest success rate
- ISS and LEO orbits outperform GTO
- Success rate improved from 0% (2006) to >80% (2020)
- Grid fins, legs, and higher block numbers correlate with success
- SVM achieved best prediction accuracy (~90%)

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!